Nathaniel Imel

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Education

New York University New York, NY
Ph.D. in Psychology (Cognition and Perception) 2024–now

University of California, Irvine Irvine, CA
Ph.D. Program in Language Science 2023–2024
Ph.D. Program in Logic and Philosophy of Science 2022–2023

University of Washington Seattle, WA
M.S. in Computational Linguistics 2020–2022

University of California, San Diego La Jolla, CA
B.A. in Philosophy 2016–2020

Experience

Basis Research Institute New York, NY
Research Intern Summer 2026

Posh Technologies Boston, MA
NLP Research Intern Summer 2021

Publications

Journal articles and top-tier proceedings

[1] N. Imel and N. Zaslavsky. Evolution and compression in LLMs: On the emergence of human-aligned categorization. In: *Proceedings of the International Conference on Learning Representations (ICLR) 2026*. Rio de Janeiro, Brazil, 2026. https://openreview.net/forum?id=s7gSTR2AqA&.

[2] N. Imel, Q. Guo, and S. Steinert-Threlkeld. An efficient communication analysis of modal typology. *Open Mind* (2026). https://doi.org/10.1162/OPMI.a.313.

[3] S. Steinert-Threlkeld, N. Imel, and Q. Guo. A semantic universal for modality. *Semantics and Pragmatics* (2023). https://doi.org/10.3765/sp.16.1.

Peer reviewed proceedings and workshop papers

[4] N. Imel, S. Chung, S. Zheng, D. Peters, T. A. Langlois, and N. Zaslavsky. Semantic efficiency and human-alignment in multilingual multimodal large language models. In: *9th Annual Conference on Computational Cognitive Neuroscience (CCN 2026)*. 2026. https://openreview.net/forum?id=rchivz9Xjd.

[5] N. Imel and N. Zaslavsky. Culturally transmitted color categories in LLMs reflect a learning bias toward efficient compression. In: *CogInterp: Interpreting Cognition in Deep Learning Models (CogInterp @ NeurIPS)*. Best paper award. 2025. https://arxiv.org/abs/2509.08093.

[6] N. Imel, J. Culbertson, S. Kirby, and N. Zaslavsky. Iterated language learning is shaped by a drive for optimizing lossy compression. In: *Proceedings of the 46th Annual Meeting of the Cognitive Science Society*. 2025. https://escholarship.org/uc/item/63d7n4v0.

[7] M. Taliaferro, N. Imel, N. Zaslavsky, and E. Blanco-Elorrieta. Bilinguals exhibit semantic convergence while maintaining near-optimal efficiency. In: *Proceedings of the 46th Annual Meeting of the Cognitive Science Society*. 2025. https://escholarship.org/uc/item/4128j529.

[8] N. Imel, C. Haberland, and S. Steinert-Threlkeld. The Unnatural Language Toolkit (ULTK). In: *Proceedings of the Society for Computation in Linguistics*. 2025. https://doi.org/10.7275/scil.3144.

[9] N. Imel and N. Zaslavsky. Optimal compression in human concept learning. In: *Proceedings of the 46th Annual Meeting of the Cognitive Science Society*. 2024. https://escholarship.org/uc/item/7pc1g61d.

[10] N. Imel and Z. Hafen. Citation-similarity relationships in astrophysics. In: *AI for Scientific Discovery: From Theory to Practice Workshop (AI4Science @ NeurIPS)*. 2023. https://openreview.net/pdf?id=mISayy7DPI.

[11] N. Imel, R. Futrell, M. Franke, and N. Zaslavsky. Noisy population dynamics lead to efficiently compressed semantic systems. In: *NeurIPS Workshop on Information-Theoretic Principles in Cognitive Systems Workshop (InfoCog @ NeurIPS)*. 2023. https://openreview.net/pdf?id=9oEXdXmMNr.

[12] N. Imel. The evolution of efficient compression in signaling games. In: *Proceedings of the 45th Annual Meeting of the Cognitive Science Society*. 2023. https://escholarship.org/uc/item/5dr5h4q0.

[13] N. Imel and S. Steinert-Threlkeld. Modals in natural language optimize the simplicity/informativeness trade-off. In: *Proceedings of Semantics and Linguistic Theory (SALT 32)*. 2022. https://doi.org/10.3765/salt.v1i0.5346.

[14] Q. Guo, N. Imel, and S. Steinert-Threlkeld. A Database for Modal Semantic Typology. In: *Proceedings of the 4th Workshop on Research in Computational Linguistic Typology and Multilingual NLP*. Seattle, Washington: Association for Computational Linguistics, July 2022. https://aclanthology.org/2022.sigtyp-1.6/.

Works in progress

[15] N. Imel, R. Futrell, M. Franke, and N. Zaslavsky. Agent-based imitation dynamics can yield efficiently compressed population-level vocabularies. (2026). (under review). https://arxiv.org/abs/2603.15903.

[16] N. Imel and N. Zaslavasky. On convexity and efficiency in semantic systems. (2026). Computational Modeling Prize in Language at CogSci 2026. https://arxiv.org/abs/2602.08238.

[17] N. Imel and Z. Hafen. Density, asymmetry and citation dynamics in scientific literature. (2025). https://arxiv.org/abs/2506.23366.

[18] W. Uegaki, A. Mucha, N. Imel, and S. Steinert-Threlkeld. Deontic priority in the lexicalization of impossibility modals. (2023). https://psyarxiv.com/h63y9/.

Talks and Presentations

On convexity and efficiency in semantic systems

Cogsci 2026 (Rio de Janeiro, Brazil) — 7/26

Culturally transmitted color categories in LLMs reflect a learning bias toward efficient compression.

CogInterp @ NeurIPS 2025 (San Diego, CA) — 12/07/25
EvoLang 2026 (Plovdiv, Bulgaria) — 04/07/26
ICLR 2026 (Rio de Janeiro, Brazil) — 04/25/26

Iterated learning is shaped by a drive for optimizing lossy compression

Cogsci 2025 (San Francisco, CA) 8/01/25

NYU C&P Miniconvention (New York, NY) 9/26/25

EvoLang 2026 (Plovdiv, Bulgaria) 04/07/26

Optimal compression in human concept learning

Cogsci 2024 (Rotterdam, NL) 7/24/24

PhilLab at Dartmouth College (virtual) 11/12/24

Noisy population dynamics lead to efficiently compressed semantic systems

University of Tübingen Linguistics Colloquium (virtual) 7/04/23

MIT Computational Psycholinguistics Lab (Cambridge, MA) 10/17/23

Evolang 2024 (Madison, WI) 5/19/24

Citation-similarity relationships in astrophysics literature

Santa Fe Institute Workshop on Intelligence and Representation (Cambridge, UK) 8/18/23

Modals in natural language optimize the simplicity/informativeness trade-off

Semantics and Linguistic Theory (Mexico City, CDMX) 6/08/22

Experiments in Linguistic Meaning (Philadelphia, PA) 5/18/22

Awards

Computational Modeling Prize in Language @ CogSci 2026 2026

Best paper award at CogInterp Workshop @ NeurIPS 2025 2025

Santa Fe Institute Complexity GAINs Summer Fellowship 2023

North American Summer School for Logic, Language and Information Student Fellowship (USC) 2023

Merit Fellowships (UC Irvine School of Social Sciences) 2022

Accepted to Summer School in Logic and Formal Epistemology (CMU) 2021

Best paper "Desire Semantics", selected for UC San Diego undergraduate philosophy journal *Intuitions* 2020

Teaching

Grader

- ANOVA (NYU) – w. Ajua Duker Fall 2025

Teaching Assistant

- Computational Cognitive Modeling (NYU) – w. Todd Gureckis Spring 2026
- Lab in Cognition and Perception (NYU) – w. Noga Zaslavsky Spring 2025
- Introduction to Syntax (UCI) – w. Galia Bar-Sever Winter, Spring 2024
- Introduction to Linguistics (UCI) w. Ben Mis Fall 2023
- Basic Economics I (UCI) w. Jiawei Chen Spring 2023
- Introduction to Symbolic Logic – w. Toby Meadows (UCI) Winter 2023

Service

Organizing

- Co-organizer for InfoCog Workshop @ CogSci 2025
- Organizing Committee for Society for Computation in Linguistics Conference 2023
- Co-organizer for InfoCog Workshop @ Neurips 2023
- Department Colloquium Committee (UCI Language Science) 2023

Reviewing

- Journal reviewing: JML, Mind & Language 2024-2025
- Conferences/Workshops: CogSci, InfoCog Workshop @ NeurIPS, EvoLang, California Annual Meeting of Psycholinguistics 2023–2026

Skills

- **Programming Languages:** Python, Java, C++, MATLAB
- **Libraries & Frameworks:** NumPy, PyTorch, pandas, plotnine, matplotlib, huggingface, NLTK, Hydra
- **Cloud & Computing:** high-performance computing with Condor & Slurm, Google Cloud Platform, Lambda AI
- **DevOps & Tools:** GitHub Workflows/Actions, containerization (Singularity)